USTHB
Faculty of Computer Science
L1 – SIGL
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Chapter 1: Introduction to Computer Science

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Dr. BOUCHEKIR Redouane
ticbur2023@gmail.com
tps://sites.google.com/view/algo-s2-2026/home

Computer science

A definition of computer science is the automatic processing of information.

Computer science is the field of study and practice that involves the processing of information using computer systems. It encompasses the design, development, use, and management of these systems, including computers and software, to solve problems, store, manipulate, and transmit data, as well as automate processes and facilitate communication.

Computer science covers a wide range of topics such as programming, computer networks, data security, artificial intelligence, database design, and many other related areas.

History

Early 20th Century:

Alan Turing introduces the concept of a theoretical machine capable of solving any computable problem, known as the Turing Machine. This becomes a foundational concept in computer science.

1930s-1940s:

The development of electronic computers begins with machines like the Atanasoff-Berry Computer (ABC) and the Colossus, used for specialized calculations during World War II.

1940s-1950s:

ENIAC, the first general-purpose electronic digital computer, is built.
John von Neumann's architecture concept becomes the basis for modern computer design.
The term "software" is coined by mathematician and computer scientist John W. Tukey.

1950s-1960s:

The programming language FORTRAN is developed, making it easier to write software for computers.
COBOL and LISP are also created during this time.
The first high-level programming languages are introduced.

1960s-1970s:

The invention of integrated circuits (ICs) leads to the miniaturization of computers and the emergence of the microprocessor.
UNIX operating system is developed at Bell Labs.
The ARPANET, a precursor to the internet, is established.

1980s-1990s:

Personal computers become popular with the introduction of the IBM PC and Apple Macintosh.
Graphical user interfaces (GUIs) like Windows and MacOS simplify computer usage.
The World Wide Web is invented by Tim Berners-Lee in 1989.

2000s-Present:

The 21st century sees rapid advancements in computer technology, including the rise of smartphones, cloud computing, and the Internet of Things (IoT).
Artificial intelligence (AI) and machine learning (ML) gain prominence.
Quantum computing research progresses, potentially revolutionizing computing in the future.

Recent Developments (Up to 2021):

The development of powerful supercomputers for scientific research and simulations.
Growing concerns about cybersecurity and data privacy.
Advancements in areas like deep learning, natural language processing, and autonomous vehicles.

Computer

Computer: A machine used for processing information. A computer is composed of two parts:

Hardware

This includes all the physical components of the computer, such as the central processing unit (CPU), memory (RAM), storage devices (hard drives, SSDs), input devices (keyboard, mouse), and output devices (monitor, printer). The hardware performs the actual data processing, storage, and communication tasks.

Software

This refers to the programs and instructions that run on the hardware. Software includes the operating system (e.g., Windows, macOS, Linux) and various applications (e.g., word processors, web browsers, games) that allow users to perform specific tasks, such as writing documents, browsing the internet, or running complex calculations.



Which operating system do you know?

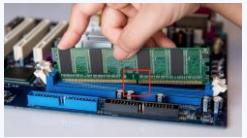
Hardware



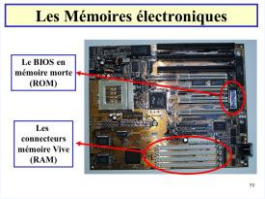
Processor (CPU)
CPU: Central Processing Unit

Memory

RAM
RAM: Random Access Memory



ROM
ROM: Read-Only Memory



Hard disk
HDD: Hard Disk Drive
SSD: Solid-State Drive



Motherboard





Graphics card
GPU: Graphics Processing Unit

Memory

- It allows the storage of data and instructions to be executed in binary form (BIT: Binary digIT).
- A bit takes 2 values: 0 and 1
- The central memory is composed of RAM (Random Access Memory), which is a random-access memory.
  RAM is volatile memory, meaning its content is lost as soon as the electrical power is turned off.
- The memory capacity refers to the number of bytes the memory holds. A byte is 8 bits.
- The central memory contains DRAM (Dynamic RAM). The access time (read time) ranges from 50 to 150 nanoseconds (ns).

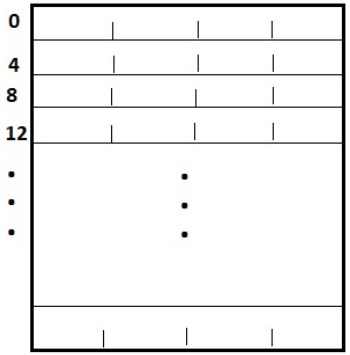
Bit: 0 ou 1 | 1 |

Byte: 8 bits

1	0	1	0	0	1	0	1

$1\ Kilobyte\ (KB)\ is\ 2^{10}\ bytes\ =\ 1024\ bytes$
$1\ Megabyte\ (MB) is\ 2^{10} KB\ =\ 1024\ KB$
$1\ Gigabyte\ (GB)\ is\ 2^{10} MB\ =\ 1024\ MB$
$1\ Terabyte\ (TB)\ is\ 2^{10} GB\ =\ 1024\ GB$

A memory word (word) is either 4 bytes (32 bits) or 8 bytes (64 bits).

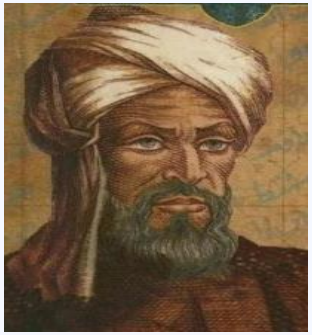


This depends on the architecture of the system

- In a **32-bit system**, a word is typically 4 bytes (32 bits).
- In a **64-bit system**, a word is typically 8 bytes (64 bits).

Algorithmics

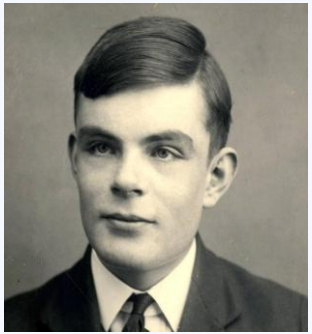
Algorithmics refers to the study and design of algorithms, which are step-by-step procedures or sets of rules used to solve problems or perform specific tasks.



Al-Khwârizmî

Algorithmics covers various topics, including:

- ***Complexity:*** Evaluating how an algorithm scales with input size (Big-O notation).
- ***Data structures:*** Organizing data efficiently to optimize algorithm performance (e.g., arrays, linked lists, trees).
- ***Optimization:*** Finding the best solution among many possible options.
- ***Recursion:*** Solving problems by breaking them down into smaller, simpler instances of the same problem.



Alan Turing

Algorithm



Problème



Steps to Problem Solving

To solve a computing problem, the following steps are typically followed:

1. Define
2. Analyze
3. Flowchart

Design

4. Translate to programming language
5. Compilation
6. V&V (Test and Debugging)

Implementation

Compilation



Program
in C

Compiler

Syntax Analysis

Semantic Analysis

Optimization

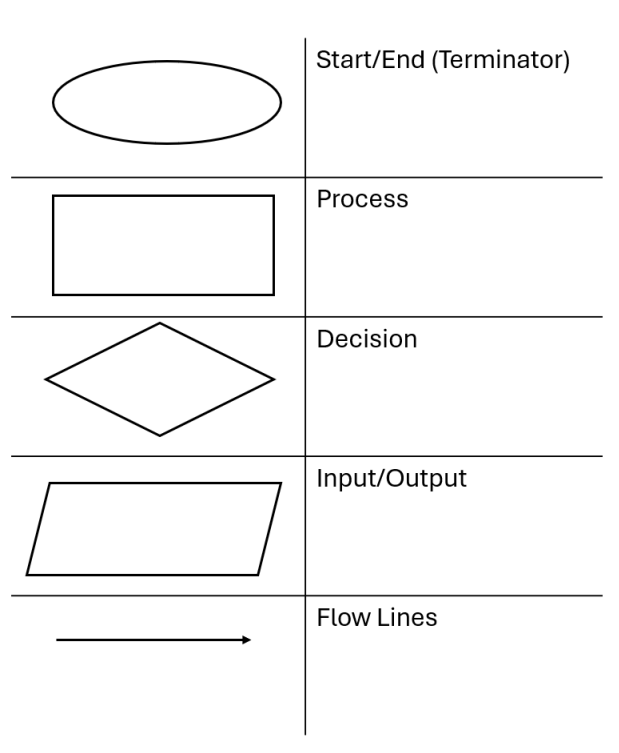
Machine Instructions

1010110110
1100101011
1000110101

Structuration: Flowchart

A flowchart is a visual representation of an algorithm or process, illustrating the sequence of steps needed to achieve a specific goal.

Flowcharts use standardized symbols to depict different types of actions or decisions within the process.

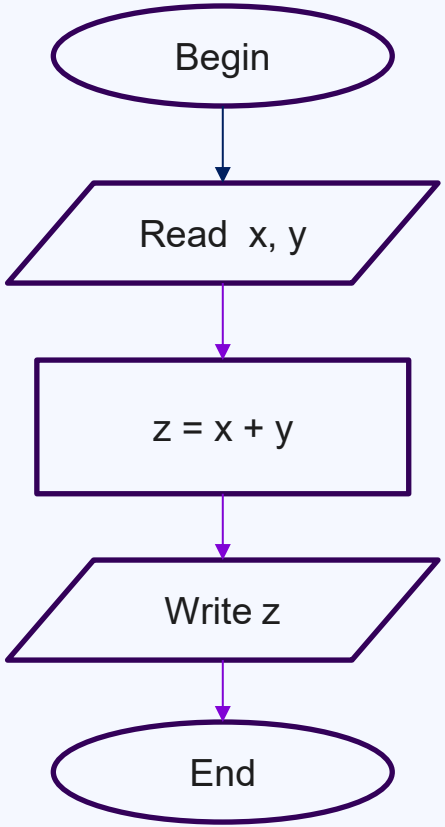


Structuration: Flowchart

Key Components of Flowcharts:

1. **Start/End (Terminator):** Represented by an oval or rounded rectangle. This symbol indicates the beginning and end of the flowchart.
2. **Process:** Depicted as a rectangle. This symbol represents a process, operation, or action that is being performed (e.g., calculations, data manipulation).
3. **Decision:** Shown as a diamond. This symbol indicates a decision point where the flow can branch based on a condition (e.g., yes/no, true/false).
4. **Input/Output:** Represented by a parallelogram. This symbol indicates input (e.g., data entry) or output (e.g., display results).
5. **Flow Lines:** Arrows connect the symbols, indicating the direction of the flow of control through the algorithm.

Example 1: Calculate the Sum of 2 Numbers



Example 2: Calculate the Average of 3 Numbers

Example 3: Calculate the Maximum of Two Numbers (Nbr1 and Nbr2)

Thanks !

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